

MATH 122 – Fall 2022

Answer Key

HW # 1: Problem # 6: a. $x = y - \frac{b}{3a}$, b. $x = y - \frac{b}{4a}$; Problem # 11: $\frac{400}{17} \approx 23.53$ feet; Problem # 12: $x = \frac{200}{9} \approx 22.22$ inches.

HW # 2: Problem # 16: a. $V = x^2y$, $S = 2x^2 + 4xy$; Problem # 21: a. slope = $2a$; Problem # 22: a. slope = $3a^2$; Problem # 29: slope = 6

HW # 3:

Drill # 49a. $(x^2 + 3x + 1)(5dx - 140x^{19}dx + 12x^{11}dx) + (5x - 7x^{20} + x^{12} - 6)(2xdx + 3dx)$; b. $(5x^2 - 7x + 1)(20x^9dx - 28x^6dx + 9x^2dx - 2dx) + (2x^{10} - 4x^7 + 3x^3 - 2x + 8)(10xdx - 7dx)$; c. $a_1dx + 2a_2xdx + 3a_3x^2dx + \dots + na_nx^{n-1}dx$; d. i. $18x^5dx - 10x^4dx + 180x^3dx$; ii. $x^4(6xdx - 2dx) + (3x^2 - 2x + 45)(4x^3dx)$

Drill # 50a. $dy = 9.8xdx + 15dx$; b. $dy = 0$; c. $dy = 0$; d. $dy = 0$; e. $dy = 2xdx - 2dx$; f. $dy = 3(3x^2dx - 4dx)$; g. $dy = 28x^3dx - 6xdx + dx$; h. $dy = 10xdx + 2dx$; i. $dy = \frac{x}{y}dx$; j. $dy = \frac{-xdx - zdz}{y}$; k. $dy = \frac{3z^2dz - y^2dx}{2xy}$; l. $dy = \frac{-yzdx - xydz}{xz}$; m. $dy = (2x - 1)(9x^2dx + 10xdx - 6dx) + (3x^3 + 5x^2 - 6x + 7)(2dx)$

Drill # 59a. $dy = 3(x^2 + 3x - 7)^2(2xdx + 3dx)$; b. $dy = \frac{1}{2}(x^2 + x)^{-\frac{1}{2}}(2xdx + dx) + 3dx$; c. $dy = -\frac{1}{4}(3(x^4 - 5x^{-3})^5 + (x^4 - 5x^{-3})^{-5})^{-\frac{5}{4}}(15(x^4 - 5x^{-3})^4 - 5(x^4 - 5x^{-3})^{-6})(4x^3dx + 15x^{-4}dx)$

Drill # 61a. $dy = dx - 4x^{-2}dx$; b. $dy = 2xdx - \frac{1}{2}x^{-\frac{1}{2}}dx$; c. $dy = \frac{2}{3}x^{-\frac{1}{3}}dx - \frac{3}{2}x^{\frac{1}{2}}dx$; d. $dy = dx - x^{-2}dx$; e. $dy = dx - x^{-2}dx$; f. $dy = \frac{1}{2}(3x^4 - 7x^3 + 3)^{-\frac{1}{2}}(12x^3dx - 21x^2dx)$; g. $dy = \frac{1}{2}\left(1 + \sqrt{1 + \sqrt{1 + x}}\right)^{-\frac{1}{2}}\left(\frac{1}{2}(1 + \sqrt{1 + x})^{-\frac{1}{2}}\left(\frac{1}{2}(1 + x)^{-\frac{1}{2}}\right)dx\right)$; h. $dy = \frac{1}{2}x^{-\frac{1}{2}}dx + \frac{1}{2}z^{-\frac{1}{2}}dz$; i. $dy = \frac{1}{2}(x + z)^{-\frac{1}{2}}(dx + dz)$; j. $dy = \frac{1}{2}(xz)^{-\frac{1}{2}}(xdz + zdz)$; k. $dy = \frac{c^2tdt - xdx - zdz}{y}$

HW # 4:

Drill # 63a. $dy = -\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}}dx$; b. $dy = \frac{3x^2dx - dx}{2y}$; c. $dy = \frac{x(x^2 + y^2)dx + 3x^2dx + y^2dx}{2y - y(x^2 + y^2) - 2xy}$; d. $dy = \frac{x(y-2)^2dx}{2y - y(y-2)^2 - (y-2)(x^2 + y^2)}$; e. $dy = \frac{2xdx - x(x^2 + y^2 + 1)dx}{y(x^2 + y^2 + 1)}$; f. $dy = \frac{2x(x^2 + 2y - 1)dx + xy^2dx}{y(1 - x^2) - 2(x^2 + 2y - 1)}$; g. $dy = \frac{x^3dx - x(x^2 + y^2)dx}{y(x^2 + y^2)}$; h. $dy = \frac{2y^2dx - (x+2)(x^2 + y^2)dx - x(y^2 + x(x+4))dx}{y(x^2 + y^2) + y(y^2 + x(x+4)) - 4xy}$

Drill # 67a. $dy = -x^{-2}dx$; b. $dy = \frac{dx}{(x+1)^2}$; c. $dy = dx - x^{-2}dx$;

d. $dy = \frac{(x^{-1} + 7x^{15})(3x^2dx + 4xdx) - (x^3 + 2x^2)(-x^{-2}dx + 105x^{14}dx)}{(x^{-1} + 7x^{15})^2}$; e. $dy = \frac{(y^2 + 1)dx - 2y(y^2 + 1)^2dx}{(y^2 + 1)^2\left(2x + \frac{1}{2}y^{-\frac{1}{2}}\right) + 2xy}$

f: $dy = \frac{2xdx + (y+1)^{-1}dx + xy(x+z)^{-2}(dx+dz) - (x+z)^{-1}ydx}{(x+z)^{-1}x + x(y+1)^{-2}}$; g: $dy = \frac{ydx}{2(1-x)}$;

h: $dy = -(x^2 - 3)(3x^4 + 6x + 2)^{-2}(12x^3dx + 6dx) + (3x^4 + 6x + 2)^{-1}(2xdx)$; i: $dy = \frac{\left(x^{\frac{1}{2}+2x^{-\frac{1}{2}}}\right)[(3x^2+2x-1)(32x^7dx+9x^{-10}dx)+(4x^8-x^{-9}+2)(6xdx+2dx)]-(3x^2+2x-1)(4x^8-x^{-9}+2)\left(\frac{1}{2}x^{-\frac{1}{2}}dx-x^{-\frac{3}{2}}dx\right)}{\left(x^{\frac{1}{2}+2x^{-\frac{1}{2}}}\right)^2}$

Problem # 72c. $y - a^2 = 2a(x - a)$; Problem # 79a. $y + \frac{15}{2\sqrt{13}} = \frac{2}{3}\left(x - \frac{5}{\sqrt{13}}\right)$, $y - \frac{15}{2\sqrt{13}} = \frac{2}{3}\left(x + \frac{5}{\sqrt{13}}\right)$;
b. $y - \frac{4}{\sqrt{2}} = -\frac{4}{3}\left(x - \frac{3}{\sqrt{2}}\right)$, $y + \frac{4}{\sqrt{2}} = -\frac{4}{3}\left(x + \frac{3}{\sqrt{2}}\right)$; c. There are no lines

HW # 5:

Problem # 82a. At $(1, 3\sqrt{3})$, tangent: $y - 3\sqrt{3} = -\sqrt{3}(x - 1)$, normal: $y - 3\sqrt{3} = \frac{1}{\sqrt{3}}(x - 1)$, At $(-1, -3\sqrt{3})$, tangent: $y + 3\sqrt{3} = -\sqrt{3}(x + 1)$, normal: $y + 3\sqrt{3} = \frac{1}{\sqrt{3}}(x + 1)$; b: At $(0, 1)$, tangent: $y - 1 = -\frac{1}{2}(x - 0)$, normal: $y - 1 = 2(x - 0)$, : At $(0, -1)$, tangent: $y + 1 = \frac{1}{2}(x - 0)$, normal: $y + 1 = -2(x - 0)$; c: At $(0, 2)$, tangent: $y - 2 = -1(x - 0)$, normal: $y - 2 = 1(x - 0)$, At $(0, -2)$, tangent: $y + 2 = 1(x - 0)$, normal: $y + 2 = -1(x - 0)$; d: At $(1, 1)$, tangent: $y - 1 = \frac{1}{3}(x - 1)$, normal: $y - 1 = -3(x - 1)$, At $(3, 3)$, tangent: $y - 3 = -\frac{1}{5}(x - 3)$, normal: $y - 3 = 5(x - 3)$;

e: tangent: $y - \frac{2^{\frac{1}{4}}}{2} = \left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)\left(x - \frac{2^{\frac{1}{4}}}{2}\right)$, normal: $y - \frac{2^{\frac{1}{4}}}{2} = \left(\frac{-\sqrt{2}-1}{\sqrt{2}-1}\right)\left(x - \frac{2^{\frac{1}{4}}}{2}\right)$; f: tangent: $y - \frac{(12+3\sqrt{3})}{26} = \frac{\frac{1}{2}\left(\frac{12+3\sqrt{3}}{26}\right)^2 + \frac{12+3\sqrt{3}}{13} - \frac{3}{4}}{\frac{3}{4}\left(\frac{12+3\sqrt{3}}{26}\right) - 2\left(\frac{12+3\sqrt{3}}{13} - \frac{3}{4}\right)}\left(x - \frac{1}{2}\right)$, normal: $y - \frac{(12+3\sqrt{3})}{26} = -\frac{\frac{3}{4}\left(\frac{12+3\sqrt{3}}{26}\right) - 2\left(\frac{12+3\sqrt{3}}{13} - \frac{3}{4}\right)}{\frac{1}{2}\left(\frac{12+3\sqrt{3}}{26}\right)^2 + \frac{12+3\sqrt{3}}{13} - \frac{3}{4}}\left(x - \frac{1}{2}\right)$; g: at $(2, 2\sqrt{3})$, tangent: $y - 2\sqrt{3} = \frac{7}{\sqrt{3}}(x - 2)$, normal: $y - 2\sqrt{3} = -\frac{\sqrt{3}}{7}(x - 2)$; at $(2, -2\sqrt{3})$, tangent: $y + 2\sqrt{3} = -\frac{7}{\sqrt{3}}(x - 2)$, normal: $y + 2\sqrt{3} = \frac{\sqrt{3}}{7}(x - 2)$; at $(-2, 2\sqrt{3})$, tangent: $y - 2\sqrt{3} = -\frac{7}{\sqrt{3}}(x + 2)$, normal: $y - 2\sqrt{3} = \frac{\sqrt{3}}{7}(x + 2)$; at $(-2, -2\sqrt{3})$, tangent: $y + 2\sqrt{3} = \frac{7}{\sqrt{3}}(x + 2)$, normal: $y + 2\sqrt{3} = -\frac{\sqrt{3}}{7}(x + 2)$; h: at $(-1, -1)$, tangent: $y + 1 = \frac{1}{2}(x + 1)$; normal: $y + 1 = -2(x + 1)$; at $(-1, 1)$, tangent: $y - 1 = -\frac{1}{2}(x + 1)$; normal: $y - 1 = 2(x + 1)$;

Problem # 87a: at $x = 0$, tangent: $y = -2$; at $x = 8$, tangent: $x = 8$; b: at $x = 0$, tangent: $x = 0$, at $x = 8$, tangent: $y = 2(x - 8)$

Problem # 88: $B = 1$, $C = 7000$; Problem # 90a: $dy = 3Ax^2dx + 2Bxdx + Cdx$; d. $A = -\frac{2H}{L^3}$, $B = \frac{3H}{L^2}$

HW # 6:

Problem # 95a. iii: the ball will go 178.5 meters ; iv: time to impact is approx. 11.036 seconds; v: impact velocity is approx. 59.153 meters/sec ; b. ii: $\frac{v_0^2}{19.6}$;

Drill # 99a. $\frac{d^2y}{dx^2} = 36x^2 - 6x$; b. $\frac{d^2y}{dx^2} = -\frac{1}{4}y^{-3}$; c. $\frac{d^2y}{dx^2} = -\frac{1}{4}x^{-\frac{3}{2}}$; d. $\frac{d^2y}{dx^2} = \frac{2y}{x}$; e. $\frac{d^2y}{dx^2} =$

$(3y - 4x - y^2)(3y^2 + 3x - 2xy + 1)^{-2} \left[(6y - 2x) \frac{dy}{dx} + 3 - 2y \right] + (3y^2 + 3x - 2xy + 1)^{-1} \left[4 + (2y - 3) \frac{dy}{dx} \right] = (3y - 4x - y^2)(3y^2 + 3x - 2xy + 1)^{-2} [(6y - 2x)(4x - 3y + y^2)(3y^2 + 3x - 2xy + 1)^{-1} + 3 - 2y] + (3y^2 + 3x - 2xy + 1)^{-1} [4 + (2y - 3)(4x - 3y + y^2)(3y^2 + 3x - 2xy + 1)^{-1}]$; Problem # 102a. speeding up; b. slowing down; c. slowing down; d. speeding up

Problem # 106b: Largest vertical acceleration is $\frac{d^2y}{dt^2} \Big|_{x=0} = \frac{6Hv^2}{L^2}$, smallest vertical acceleration is

$\frac{d^2y}{dt^2} \Big|_{x=L} = -\frac{6Hv^2}{L^2}$; d: $L \geq 100^2 \cdot \sqrt{60} \approx 77.459 \text{ km}$.

Problem # 107a: $A = -\frac{9.8}{180^2}$; b: $x = \frac{180^2}{9.8} \approx 3306.12 \text{ m} \approx 3.31 \text{ km}$; c: $t = \frac{180\sqrt{2}}{9.8} \approx 25.97 \text{ sec.}$;

d: $\frac{ds}{dt} = \sqrt{\left(\frac{180}{\sqrt{2}}\right)^2 + \left(-9.8t + \frac{180}{\sqrt{2}}\right)^2} = \frac{180}{\sqrt{2}} \sqrt{1 + \left(1 - \frac{19.6x}{180^2}\right)^2}$